

Year 8 Science

Earth and Space Sciences

Investigation - Rate of Cooling

INVESTIGATION 9.2

Does fast cooling make a difference?

AIM To investigate the effect of cooling rate on the size of crystals

Materials:

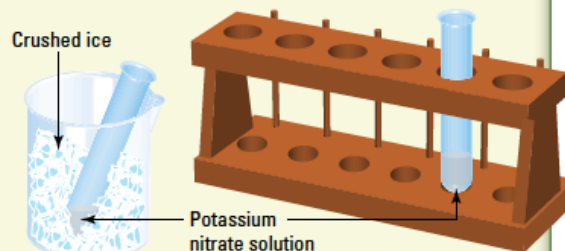
freshly made saturated solution of potassium nitrate
potassium nitrate
spatula
250 mL beaker
3 test tubes and test-tube rack
test-tube holder
Bunsen burner, heatproof mat and matches
crushed ice
safety glasses
hand lens

CAUTION Safety glasses must be worn during this experiment.

METHOD AND RESULTS

- ▶ Half-fill a beaker with crushed ice.
- ▶ Quarter-fill a clean test tube with saturated potassium nitrate solution. Add a spatula of potassium nitrate.
- ▶ Gently heat the solution over a Bunsen burner flame until the added potassium nitrate has dissolved or until the solution starts to boil.
- ▶ Pour half the warm solution into each of two clean test tubes.

- ▶ Place one test tube in the beaker of crushed ice and the other test tube in the rack to cool.
- ▶ When crystals have formed in each test tube, examine them with a hand lens.



Cool one solution quickly and the other one slowly.

- 1 Draw a labelled diagram of some crystals in each test tube, concentrating on their shape and size.

DISCUSS AND EXPLAIN

- 2 Which test tube contained the larger crystals — the one that cooled quickly or the one that cooled slowly?
- 3 Which type of extrusive rock would you expect to have the larger crystals — those that cool slowly on the surface or those that cool quickly underwater?
- 4 Why do safety glasses need to be worn during this experiment?